

Weijie (Jay) CAO

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SUMMARY

Data analyst with an M.Sc. in Mathematical Finance & FinTech and 2+ years across fintech, quant, and HR analytics. I turn messy multi-source data into reliable SQL pipelines, BI dashboards, and statistical / ML models that drive decisions.

EXPERIENCE

Holistic Index — Data Analyst

Sep 2024 – May 2026

Chicago, IL (Remote)

- Designed and maintained SQL data pipelines integrating ATS/HRIS sources, safeguarding the accuracy and security of sensitive HR data (compensation, performance, attrition).
- Built and shipped ~40 automated Looker Studio dashboards per year tracking retention, time-to-fill, and hiring-funnel metrics, improving reporting turnaround and visibility.
- Partnered with client managers to streamline reporting workflows, cutting manual data-prep time and accelerating data-driven decisions.

Trade Terminal — Quant Researcher Intern

Feb 2024 – Sep 2024

San Jose, CA

- Wrote SQL for efficient retrieval across large market datasets, supporting in-depth quantitative research and model development.
- Back-tested investment strategies in Python, improving portfolio risk-adjusted returns through data-driven analysis and optimization.

Springer Capital — Financial Analyst Intern

Jun 2023 – Aug 2023

Wilmette, IL

- Conducted due diligence comparing occupancy and average room rates across five comparable hotel properties; cleaned and transformed raw data into a model to forecast investment returns.

Allied Millennial Partners, LLC — Data Analyst Intern

Feb 2021 – Apr 2021

New York, NY

- Wrangled and modeled data in Python (NumPy, Pandas), including ARIMA forecasts; ran diagnostics (heteroskedasticity, autocorrelation, KS / Q-Q) and visualized results in R (ggplot2).

PROJECTS

Gravity (心引力) — AI Emotion-Driven Social App (full-stack)

github.com/WeijieCao77/lingzhi_comp • [live demo](#)

- Used an LLM to parse free-text / voice mood into a structured emotion vector (valence × arousal), then matched users via nearest-neighbor search in emotion space with explainable similarity scores.
- Built a provider-agnostic OpenAI / Claude adapter with graceful local fallback; full-stack (backend + frontend), deployed on Railway.

Voice Visitor-Registration Agent — realtime telephony AI agent

github.com/WeijieCao77/jingyu-task

- Built a real-time Chinese voice agent that answers inbound calls, collects structured data (plate / company / phone / purpose), and pushes it to security for remote gate release in ≤25 seconds.
- LiveKit Agents + OpenAI realtime speech-to-speech (swappable STT→LLM→TTS pipeline); Twilio SIP telephony, Postgres-backed query assistant.

EDUCATION

Boston University — Questrom School of Business, Boston, MA

Sep 2022 – Jan 2024

M.Sc. Mathematical Finance & Financial Technology | GPA 3.54 / 4.00 | Questrom Dean's Achievement Scholarship

Boston University — College of Arts & Sciences, Boston, MA

Jan 2019 – Jun 2022

B.A. Mathematics & Statistics, Minor in Computer Science | Coursework: Data Science in R, Statistics, Computing Systems, Geometric Algorithms

SKILLS

Analytics & Modeling: SQL, statistical analysis & hypothesis testing, time-series (ARIMA), regression, A/B testing, machine learning.

Programming & Tools: Python (Pandas, NumPy, scikit-learn), R (ggplot2), C++, MATLAB, JMP, LaTeX.

Data & BI: Looker Studio dashboards, ETL / data pipelines, ATS/HRIS & financial data, Excel modeling, Postgres/SQLite.

Certification & Languages: FRM Part I | Mandarin (native), English (fluent).